

LINUX ASSIGNMENT-1

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SECTION:B,ROLL NO:21

QUESTION 1:

The Linux OS is a free, community-based, and open-source software created from the Unix template, which operates the same way as Windows and Mac by running the system resources, software, and hardware simultaneously. Unlike proprietary operating software, Linux can be changed, customized, and distributed without any fees. It boasts an impressive array of mobile system, server, desktop, and even some embedded system functionalities.

The pros of Linux are

- 1) Open-source and Free. Linux can be downloaded, used, and modified without any pay and entire licensing is not imposed.
- 2) Secure and Stable. Linux is much more difficult to hack and is much more resistant to crashing than the majority of operating software.
- 3) Customizable. Linux allows the user to change the kernel and desktop environments as they see fit to improve the users experience.

The cons of Linux are

- 1) Software Compatibility. Unfortunately, several applications and games are not compatible.
- 2) Learning Curve. The system is user unfriendly as command line functionality and system administration is unclear.
- 3) Hardware Support Deficiencies. There are some devices that lack official drivers, such as printers and Wi-Fi cards.

QUESTION 2:

1. Linux

* Open Source - Completely free and open source (Ubuntu, Fedora, Debian, etc. Many distributions)

QUESTION 4:

Structure Relating to File Systems

Linux adopts a “bottom-up” approach as it shapes every directory and file folder extending from a singular root directory. All Linux files and folders originate from root which is portrayed as a /. Unlike Windows which consists of both C and D Drives, Linux allows for the mounting of a singular root directory.

Basic Directories Traceable from the Root Directory Slash:

- /bin--> Necessary files for user binaries such as ls, cp, and mv.
- /boot--> Files for the bootloader which includes the kernel, initrd, and GRUB, and others.
- /dev--> Files pertaining to devices such as disks, or the /dev/sda.
- /etc--> A collection of files that configures a system.
- /home--> Directories for other users.
- /lib--> Libraries that are shared and are required by binaries for both /bin and /sbin.
- /media--> A merge point for USBs, CDs, and other removable devices.
- /mnt--> A point for the temporary mounting of files and storage systems.
- opt--> Software packages that are not mandatory for a system to function.
- /proc--> A system that is virtual containing exclusive information of the process, as well as the kernel.
- /root--> Acts as the main directory for a super user.
- /sbin--> Important binaries of the system such as shutdown and reboot.
- /tmp--> Files that are temporary and are deleted during a system restart.
- /usr--> Contains programs, documentation, and libraries for users as well as other files such as /usr/bin, and /usr/lib.
- /var--> A dynamic system containing files that are logs, cache, spool mail, and other variable files.

/bin -> Important programs stored as files

/boot -> Files of the boot loader and kernel

/dev -> Files of the devices

/etc -> Files of settings

/home -> Directories of a registered user

/lib -> Library files

/media -> Storage devices which can be removed

/mnt -> Mounting filesystems for temporary use

/opt -> Optional

QUESTION 5:

open source means you are provided with the source code at no cost (with associated licensing agreements). It does not mean: that every part (documentation, certifications, integrations, services) is free, or that customers will want to run mission-critical systems on something that has no support, or that enterprises will want to use anything that has no guarantees.

Red Hat's Business Structure: Their Monetization Strategy

Here are the key factors contributing to Red Hat's business structure. Many of these factors can also apply to other organizations in the open - source commercial ecosystem.

Subscriptions / Support / Maintenance: The subscriptions provide customers with the ability to receive ongoing updates of software, patches for security vulnerabilities, confirmed bug fixes, long-term software maintenance, certification of the hardware/software stack, interoperability with hardware/software, access to support (including SLAs), etc.

Lifecycle Management, stability guarantees: Enterprise customers want stable, tested systems that they're certain will have predictable windows for maintenance, hardware/software compatibility, and a long supported lifecycle.

Market Size & Data: How Big and How Growing :

Red Hat has had consistent growth: in 2017 total revenue was ~US\$2.4 billion, and they projected ~US\$2.7B for next year.

The ecosystem effect is huge: IDC (backed by Red Hat) reported that for each dollar of revenue Red Hat earned in 2019, its ecosystem earned ~\$21.74. In other words, Red Hat's contribution to multiple partners, hardware, and software vendors multiply its value.

QUESTION 6:

For displaying date and time in Linux we use bash scripting

THE CODE:

Input: date

Output: Tue Sep 23 00:10:18 IST 2025

This command shows:

- Day of the week
- Month and date
- Current time(HH:MM:SS)
- Time zone
- Year

QUESTION 7:

Command to Check System Uptime in Linux:

This also uses bash for performing

Input: Uptime

Output: 00:12:45 up 5 days, 3:21, 2 users, load average: 0.15, 0.10, 0.05

- up 5 days, 3:21 → System has been running for 5 days and 3 hours 21 minutes.
- 2 users → Number of users currently logged in.
- load average → System load over the last 1, 5, and 15 minutes.

By this we will get the required system running time.

QUESTION 8:

SHUTDOWN

The more regulated and "graceful" method of shutting down the system is to use the shutdown command. This high-level command manages a number of tasks before stopping the system, including:

- 1) Sending out a warning to all users who are currently logged in to save their work.
- 2) blocking the login of new users.

The -h option, which stands for "halt," now indicates that the shutdown should take place right away. Thus, shutdown -h now is a reliable and secure method of stopping a system, guaranteeing that all processes are appropriately ended and that data is preserved.

THE HALT COMMAND:

The command to stop

Conversely, the halt command is a lower-level command that merely tells the hardware to halt all CPU operations. This command would frequently stop the system but leave the power on in older systems, necessitating a manual power-off. The behaviour of halt is nearly the same as that of shutdown -h on modern systems that use systemd because it is usually associated with the shutdown command and will also shut down.

- shutdown -h : Primary function-Graceful shutdown

key features :It warns users, stops services cleanly, it also syncs data, and powers off the system.

Best Use Case: The recommended, safe way to power off a server or multi-user system.

- Halt: Primary function- Halt CPU functions

key features: A lower-level command that directly halts the system. Behavior on modern systems is often the same as shutdown -h.

Best Use Case: Convenient for single-user systems where a graceful shutdown isn't a concern, though shutdown is still preferred.

QUESTION 9:

1) init 0:

- IT changes the **runlevel** of the system to **0**, which is the **halt state**.
- IT Immediately starts shutting down processes to reach runlevel 0.
- **Behavior:**
 - Directly tells the init system to enter runlevel 0.
 - May **not notify logged-in users**.
 - May **not give services enough time to stop gracefully**.

Essentially a “hard” way to stop the system; faster, but riskier.

2)shutdown-h:

- Tells the system to **halt safely**, optionally at a scheduled time.
- **Behavior:**
 - Notifies logged-in users (e.g., via wall)

- Stops services.

WHICH IS SAFER?

shutdown -h is safer because it performs a controlled, orderly shutdown, ensuring that services stop gracefully, filesystems are synced/unmounted, and users are notified.

Although init 0 is quicker, it can be dangerous because sudden shutdowns can cause data loss or corruption, particularly on systems with services or users logged in.

QUESTION 10:

. It can cause:

- File system corruption.
- Databases getting damaged.
- Hardware stress (especially disks).
- Data loss or incomplete transactions.

BRAIN STORMING:

a) As Linux Kernel is open-source, can we build our own operating system?

Yes! That's exactly how Ubuntu, Fedora, Arch, and others were made.

b) In order to do that, what are the stoppers, hurdles, and challenges?

- Kernel coding is complex.
- Hardware drivers need to be supported.
- Security, package manager, and GUI all must be built.
- Needs a big developer community.

c) Is anyone in India working on this field? Find at-least three to four engineers.

Yes, many Indian engineers contribute to Linux kernel development:

- Vivek Goyal – Works on Linux Kernel crash dumping (Red Hat).
- Pratyush Anand – Works on ARM architecture (Red Hat India).
- Anup Patel – Works on RISC-V Linux support.
- Aneesh Kumar K.V. – Filesystem and storage development.